EgoShift: Bounding the Action-Only Shortcut in Egocentric Spatial Updating for Vision–Language Models

Supplementary Material

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This document contains the material the main paper defers to the supplement. Section 1 gives the benchmark composition. Section 2 gives the complete model panel, the Task-B answer-axis decomposition, the Task-C per-class breakdown, and the Task-B error taxonomy. Section 3 gives the full per-rotation breakdown. Section 4 records Task A's outstanding checks. Section 5 gives the per-task human reference. Section 6 states the reporting protocol. Section 7 gives the benchmark comparison. Section 8 reports compute and cost. Section 9 is the datasheet. Section 10 states the frozen protocols for the checks that have not yet run. No number here is new. Every value comes from the same reparsed result registry and candidate build audits that produce the main paper's tables.

1. Benchmark composition

Tab. 1 gives the composition of each task. It reports how many questions are held out and how many released for development, how many rooms and object categories the held-out sets draw on, and the reference scores each task is read against.

	A: same object? (control)	**B: where after a turn?** (primary)	**C: still visible?** (supporting)
Held-out questions	5,838	1,098	5,850
Development (public)	3,582	440	3,705
Development (tuning)	3,722	592	3,720
All questions	13,142	2,130	13,275
Rooms in the held-out set	3,119	736	1,334
Object categories, held out	28	45	49
Largest category's share	28%	5%	13%
Held-out rooms seen before	0	0	0
Random guessing	50.0%	25.0%	33.3%
Always the commonest answer	50.0%	31.7%	33.3%
Action-only ceiling	n/a	45.90%	n/a
Ceiling met in every cell	n/a	yes (10/10)	n/a

Table 1. Composition of EgoShift, from the current build audits. Only Task B states an action, so only Task B has an action-only ceiling; all ten of its turn cells sit exactly on it (Table 1 of the main paper). Tasks A and C are balanced by construction, so guessing at random and always giving the commonest answer are their references. Section 5 of the main paper states what these questions have not been tested for.

2. Complete model panel

Tab. 2 lists all 36 evaluated systems. Table 2 of the main paper shows the twelve frontier configurations and four representative open-weight systems. The twenty remaining open-weight models appear only here. Both tables order models by Task-B accuracy.

We produce every cell by reparsing the stored raw completions with one model-independent parser, rather than by trusting the parsed field each provider wrote. That matters in at least one case. InternVL3-38B on Task C moves from a stored 56.8% to 65.2% under the shared parser, the largest single correction in the panel. Thirty-nine duplicate keys across three cells disagree between valid retries. We count those questions incorrect, as we do parse failures, rather than keeping whichever completion scores better.

Tab. 3 decomposes every Task-B score into accuracy on lateral gold, accuracy on sagittal gold, and the fraction of the model's own answers that are lateral. The main paper cites four rows from this table. The decomposition separates a model that is uncertain from one that has adopted a fixed answer convention. A system that answers laterally on 99% of items cannot score above the lateral gold rate of 47.3%, however good its perception is. That is why eleven of 36 models sit below the nominal 25% four-way rate without behaving randomly.

Tab. 4 gives the same per-class treatment to Task C: each model's accuracy on visible, out-of-view, and occluded gold on the frozen 1,200-item subset, whose class prior is exactly one third. The main paper reports Task-C classes separately rather than pooled, and this table is that report for

Model	A	B	C	Model	A	B	C
Gemini 3 Pro	91.8	81.9†	88.2†	Qwen3-VL-8B	92.2	41.3	63.5
Gemini 3 Pro (thinking)	91.4	83.6	88.0	SenseNova-SI-1.1-Qwen3-VL-8B	88.8	36.9	64.2
Gemini 3 Flash	96.6	83.5	89.3	SenseNova-SI-1.3-Qwen3-VL-8B	93.6	33.3	61.7
Claude Fable 5	94.8	81.4	88.4	InternVL3-2B	53.9	28.8	39.7
GPT-5.6 Sol (thinking)	93.2	81.1	88.0	SenseNova-SI-1.1-Qwen2.5-VL-3B	50.0	27.9	30.8
Claude Opus 5 (extended)	95.1	80.2	84.3	InternVL3-8B	79.2	26.9	60.9
Claude Opus 5	91.6	80.1	83.2	LLaVA-OneVision-7B	50.3	25.6	39.2
Claude Sonnet 5	92.6	77.1	75.8	Idefics3-8B	59.2	24.6	37.8
GPT-5.6 Sol	93.4	76.3	87.2	SenseNova-SI-1.1-Qwen2.5-VL-7B	56.8	23.9	56.2
InternVL3-38B	96.1	65.1	65.2	Qwen2.5-VL-3B	54.4	23.3	38.7
InternVL3-78B	91.9	56.1	63.6	EgoMind-7B	59.6	21.9	33.4
Gemini 3 Flash Lite	96.2	53.0	67.3	SmolVLM-Instruct	51.8	21.4	33.4
Qwen3-VL-32B	95.6	47.7	59.3	Molmo2-8B	71.1	21.3	52.0
Qwen2.5-VL-32B	80.5	46.4	59.9	LLaVA-1.6-Mistral-7B	49.3	21.2	33.3
InternVL3-14B	92.0	45.6	63.9	Qwen2-VL-2B	50.3	19.1	32.9
Qwen2.5-VL-72B	85.7	45.6	61.8	Qwen2.5-VL-7B	60.8	18.9	41.4
Claude Haiku 4.5	71.2	45.4	38.8	Cosmos-Reason1-7B	76.1	15.8	41.4
GPT-5.1	90.8	42.0†	73.1	Qwen2-VL-7B	50.0	11.7	62.2
Nominal random	50.0	25.0	33.3	Task-B action-only ceiling	–	45.9	–

Table 2. Complete model panel, all 36 evaluated systems, ordered by Task-B accuracy. This is the full version of Table 2 in the main paper, which shows the twelve frontier configurations and four representative open-weight systems. Task A is scored on the frozen 1,192-item subset, Task B on the complete 1,098-item held-out pool, and Task C on the frozen 1,200-item subset, so every model is scored on identical items. Parse failures count as incorrect. †Three cells carry duplicate keys that disagree between valid retries. Those questions count incorrect, matching Table 2. We never resolve a conflict by keeping the completion that scores better.

Model	Acc	Lat	Sag	%Lat	Model	Acc	Lat	Sag	%Lat
Gemini 3 Pro (thinking)	83.6	92.7	75.5	56.6	Qwen3-VL-8B	41.3	68.0	17.4	78.9
Gemini 3 Flash	83.5	92.9	75.1	57.0	SenseNova-SI-1.1-Qwen3-VL-8B	36.9	74.4	3.3	94.3
Gemini 3 Pro	81.9	90.8	73.9	56.3	SenseNova-SI-1.3-Qwen3-VL-8B	33.3	70.5	0.0	99.6
Claude Fable 5	81.4	92.7	71.3	58.8	InternVL3-2B	28.8	22.7	34.2	27.5
GPT-5.6 Sol (thinking)	81.1	92.3	71.2	58.6	SenseNova-SI-1.1-Qwen2.5-VL-3B	27.9	44.1	13.3	78.1
Claude Opus 5 (extended)	80.2	91.3	70.3	58.8	InternVL3-8B	26.9	14.8	37.7	25.9
Claude Opus 5	80.1	90.8	70.6	58.4	LLaVA-OneVision-7B	25.6	47.4	6.0	92.2
Claude Sonnet 5	77.1	91.7	64.1	61.7	Idefics3-8B	24.6	13.3	34.7	18.7
GPT-5.6 Sol	76.3	91.5	62.7	62.8	SenseNova-SI-1.1-Qwen2.5-VL-7B	23.9	29.9	18.5	78.7
InternVL3-38B	65.1	89.0	43.7	72.0	Qwen2.5-VL-3B	23.3	48.4	0.9	98.2
InternVL3-78B	56.1	91.1	24.7	80.1	EgoMind-7B	21.9	37.4	8.1	88.7
Gemini 3 Flash Lite	53.0	89.4	20.4	84.2	SmolVLM-Instruct	21.4	11.0	30.7	47.0
Qwen3-VL-32B	47.7	91.1	8.8	91.4	Molmo2-8B	21.3	1.5	39.0	1.6
Qwen2.5-VL-32B	46.4	97.5	0.7	99.3	LLaVA-1.6-Mistral-7B	21.2	0.4	39.9	0.5
InternVL3-14B	45.6	82.1	13.0	84.2	Qwen2-VL-2B	19.1	8.5	28.7	35.6
Qwen2.5-VL-72B	45.6	95.8	0.7	98.1	Qwen2.5-VL-7B	18.9	37.4	2.4	96.9
Claude Haiku 4.5	45.4	94.6	1.2	98.1	Cosmos-Reason1-7B	15.8	22.2	10.0	84.1
GPT-5.1	42.0	84.4	4.0	93.1	Qwen2-VL-7B	11.7	23.7	1.0	98.5

Table 3. Task-B answer-axis decomposition on the complete 1,098-item held-out pool. **Acc** is overall accuracy; **Lat** and **Sag** are accuracy on lateral (left/right) and sagittal (front/behind) gold; **%Lat** is the fraction of the model's own answers that are lateral, against a gold lateral rate of 47.3%. Models whose %Lat departs far from 47.3 answer on one axis regardless of the image: Qwen2.5-VL-32B and -72B answer laterally on 99.3% and 98.1% of items and score 0.7% on sagittal gold, while Molmo2-8B and LLaVA-1.6-Mistral-7B answer laterally on 1.6% and 0.5%. This is what places eleven of 36 models below the nominal 25% four-way rate.

the model side. It shows that the occluded class is where Task C discriminates. The human panel behaves the same way, at 74.4% on occluded questions against roughly 97% on the other two.

Tab. 5 assigns every incorrect Task-B answer to one category. The concentration is what turns the failure mode from a description into a finding. For the eight strongest frontier configurations, between 92.0% and 100.0% of all errors are the single assumed-ahead substitution. The residual category absorbs everything the two named modes do not explain, and it never exceeds 8.0% for any frontier system.

Fig. 1 shows two items where the substitution is visible directly. The two items are mirror images of one another, and the errors mirror too. That is the clearest evidence that the effect is a property of how the models answer rather than a sign convention in our geometry.

3. Task-B accuracy by action cell

Tab. 6 is the full version of the profile in Figure 3 of the main paper, reporting every configuration and the assumed-ahead rule for each rather than the two cited in the text. Every configuration has both arms, so no cell is imputed.

Model	Vis	OoV	Occ	Model	Vis	OoV	Occ
Gemini 3 Flash	99.5	97.0	71.5	SenseNova-SI-1.3-Qwen3-VL-8B	69.5	100.0	15.5
Claude Fable 5	97.0	80.2	88.0	InternVL3-8B	81.2	97.5	4.0
Gemini 3 Pro	98.8	99.0	66.8	Qwen2.5-VL-32B	65.8	99.0	15.0
Gemini 3 Pro (thinking)	98.8	98.8	66.5	Qwen3-VL-32B	76.0	100.0	2.0
GPT-5.6 Sol (thinking)	98.5	92.8	72.8	SenseNova-SI-1.1-Qwen2.5-VL-7B	62.2	93.8	12.8
GPT-5.6 Sol	98.2	96.0	67.2	Molmo2-8B	56.2	99.8	0.0
Claude Opus 5 (extended)	98.8	91.2	63.0	Qwen2.5-VL-7B	20.5	98.5	5.2
Claude Opus 5	99.0	90.8	60.0	Cosmos-Reason1-7B	1.0	86.5	36.8
Claude Sonnet 5	94.2	96.5	36.5	InternVL3-2B	41.0	75.8	2.2
GPT-5.1	94.8	99.8	24.8	LLaVA-OneVision-7B	68.0	11.8	38.0
Gemini 3 Flash Lite	99.0	100.0	3.0	Claude Haiku 4.5	13.2	99.5	3.5
InternVL3-38B	95.0	99.0	1.5	Qwen2.5-VL-3B	34.2	3.8	78.0
SenseNova-SI-1.1-Qwen3-VL-8B	50.2	88.2	54.0	Idefics3-8B	14.5	99.0	0.0
InternVL3-14B	86.0	97.2	8.5	SmolVLM-Instruct	100.0	0.0	0.2
InternVL3-78B	91.0	98.8	1.0	EgoMind-7B	0.0	100.0	0.2
Qwen3-VL-8B	54.2	97.0	39.2	LLaVA-1.6-Mistral-7B	0.0	100.0	0.0
Qwen2-VL-7B	88.2	81.5	16.8	Qwen2-VL-2B	96.2	2.5	0.0
Qwen2.5-VL-72B	80.8	99.8	5.0	SenseNova-SI-1.1-Qwen2.5-VL-3B	88.5	0.0	3.8

Table 4. Task-C accuracy per answer class (visible / out of view / occluded) on the frozen 1,200-item subset, 400 items per class, ordered by mean per-class accuracy. The occluded class is the discriminating stratum: only Claude Fable 5 exceeds 80% on it while most open-weight systems sit below 20%, scoring far higher on the other two. Per-class values must be read jointly: a class-biased responder scores high on its preferred class regardless of the image (Qwen2.5-VL-3B reaches 78.0% occluded while scoring 3.8% out of view), and several small models are near-degenerate single-class responders. This is the same response-convention collapse the answer-axis decomposition exposes on Task B (Tab. 3).

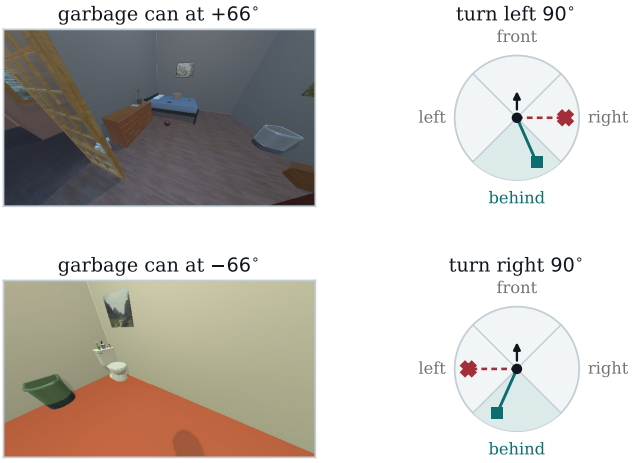


Figure 1. Two held-out questions where the five strongest frontier configurations all gave the same wrong answer. The correct answer is *behind* in both (teal); all five answered the side zone (red). The mirroring rules out a sign error in our geometry.

The assumed-ahead columns give the percentage of a model's answers that match $\mathrm{zone}(\delta)$. That is the prediction from applying the stated turn δ to a target assumed to lie straight ahead, rather than to the observed bearing. That rule agrees with the gold label about 33% of the time at 90° and 180°, and about 50% of the time in the other three cells. A model that follows it is therefore invisible at 30°, 60° and 120°, and fails almost completely at the other two. Ten of twelve configurations match it on at least 88% of 90° items.

4. Task-A provisional status

The main paper reports Task A as a positive control and makes no release claim for it. We record the outstanding checks here so the reason is on the record rather than implied. An earlier Task-A pool linked full visibility in both views almost perfectly with the *different* label. We built the current candidate with a selection intended to remove that cue, and it is balanced $50/50$ globally and within its recorded object-type and construction cells. A later continuous-caliper selection produced a different and smaller question set that has not replaced it, so the selection is not settled. The current build also reuses 26,284 frame files from its predecessor. We compared only three of them against a fresh render, so the render-equivalence audit is a spot check rather than a full one. Task A therefore stays provisional until the selection is fixed, a learned probe over the public fields is run against it, and a broader render-equivalence audit is complete. None of this affects Tasks B or C, which are built and audited independently, and no Task-A number enters a composite score.

5. Trained-human reference

Tab. 7 gives per-task human accuracy with crossed rater-by-scene bootstrap intervals and Fleiss' κ. Session counts differ by task because authorized returns exceeded the five-rater target unevenly. We retain all valid sessions rather than selecting five after observing scores. We do not release per-rater or per-item results, because the response files are the identifying records described in the main paper's human section.

Model	Errors	Ahead	Axis	Resid.
Frontier				
Gemini 3 Flash	181	100.0	0.0	0.0
Claude Fable 5	204	99.0	0.0	1.0
Gemini 3 Pro (thinking)	180	98.3	0.0	1.7
Gemini 3 Pro	199	86.9	0.0	13.1
Claude Opus 5 (extended)	217	97.2	0.0	2.8
Claude Opus 5	218	96.8	0.0	3.2
GPT-5.6 Sol (thinking)	207	95.7	0.0	4.3
GPT-5.6 Sol	260	92.7	0.0	7.3
Claude Sonnet 5	251	92.0	0.0	8.0
Gemini 3 Flash Lite	516	90.7	9.3	0.0
Claude Haiku 4.5	600	73.2	26.7	0.2
GPT-5.1	636	44.0	50.5	5.5
Frontier pooled	**3670**	**82.0**	**14.3**	**3.7**
Open-weight				
InternVL3-38B	383	77.5	0.0	22.5
Qwen2.5-VL-72B	597	72.5	25.8	1.7
Qwen2.5-VL-32B	588	72.4	27.0	0.5
Qwen3-VL-32B	574	66.2	28.7	5.1
InternVL3-78B	482	55.2	31.1	13.7
SenseNova-SI-1.3-Qwen3-VL-8B	732	48.8	50.7	0.5
Qwen3-VL-8B	644	47.8	0.0	52.2
SenseNova-SI-1.1-Qwen3-VL-8B	693	47.3	46.3	6.3
InternVL3-14B	597	47.1	37.7	15.2
SenseNova-SI-1.1-Qwen2.5-VL-7B	836	26.9	0.0	73.1
EgoMind-7B	857	26.6	70.4	3.0
LLaVA-OneVision-7B	817	25.8	70.4	3.8
Qwen2.5-VL-3B	842	24.2	74.0	1.8
SenseNova-SI-1.1-Qwen2.5-VL-3B	792	21.3	0.0	78.7
Qwen2.5-VL-7B	890	17.9	82.1	0.0
Cosmos-Reason1-7B	925	14.1	78.4	7.6
InternVL3-2B	782	12.9	0.0	87.1
SmolVLM-Instruct	863	11.6	0.0	88.4
Qwen2-VL-7B	969	10.6	88.9	0.5
Qwen2-VL-2B	888	7.9	0.0	92.1
InternVL3-8B	803	7.6	0.0	92.4
Idefics3-8B	828	7.1	80.6	12.3
LLaVA-1.6-Mistral-7B	865	4.6	94.9	0.5
Molmo2-8B	864	3.4	95.5	1.2
Open pooled	**18111**	**27.4**	**44.0**	**28.5**

Table 5. Task-B error taxonomy on the complete 1,098-item held-out pool. Every incorrect answer is assigned to exactly one category, in priority order. **Ahead**: the answer equals $\mathrm{zone}(\delta)$, the zone from applying the stated turn to a target assumed to lie straight ahead. **Axis**: the model answers on one axis for more than 80% of all items and the error lies on that axis, a response convention rather than a geometric slip. **Resid.**: everything else, including unparsed responses, under 0.3% for every model. Two named modes account for 96.3% of frontier errors and the assumed-ahead rule alone for 82.0%; for the eight strongest configurations it accounts for 86.9–100.0%. Questions with two conflicting valid responses count as errors and fall in **Resid.** The proportions invert for small open-weight systems, whose errors are dominated by answer-axis collapse.

One rater returned Tasks B and C after the freeze. The panel closed on 27 August 2026 with 25 sessions, nine for Task A and eight each for Tasks B and C. One rater who had already completed Task A returned Tasks B and C on 29 August, and we added both. Every task therefore has nine raters. On the 25-session panel mean rater accuracy was 83.8% on Task B and 90.3% on Task C, against 84.6% and 89.6% with the ninth rater included. The added sessions raise Task B by 0.8 percentage points and lower Task C by 0.7. Per-turn Task-B rater accuracy moves by at most 1.2 percentage points, and the reported gap between models and raters at $90°$ and $180°$ widens by 1.2 and 0.7. We report the nine-rater panel throughout. We retain the scored outputs for the 25-session panel so that both can be checked.

Protocol deviations in full. Annotators returned twenty-seven complete sessions, nine for each task. Every session carries 120 measured items, practice examples and three instructed-response checks. Before scoring we verified the returned item identifiers, the package identifier, the offered options and completeness against the frozen package. No session failed the frozen two-of-three instructed-response rule, and accuracy is never an exclusion criterion. We exceeded the five-rater target after additional returns and kept every valid session, rather than selecting five after their scores were visible. None of the nine raters with timestamped histories completed Task B first, against an intended B-first ordering. The Task-B estimate is therefore a trained-panel result that may carry cross-task familiarity, not a first-exposure estimate. We replaced one withdrawn Task-B session on the same items. The withdrawn file is unused, and we do not count the replacement as an additional participant. All nine Task-A sessions missed the same instructed-response item, and seven of nine Task-C sessions missed the same one. In both cases the prompt, the offered options and the key agree. We therefore record these as instrument failures, not as grounds for removing raters who meet the frozen rule. These deviations bound the scope of the reference and alter no item response or benchmark label. This cohort estimates trained task performance. It is not an item-adjudication study, and we removed or edited no item on the basis of these responses.

6. Reporting protocol

These are the rules Section 4.2 of the main paper summarises. Report Task B one turn at a time and Task C one answer class at a time. A pooled number averages groups whose answer sets differ in size, and it can move in the opposite direction to the underlying ability. Treat questions from the same room as dependent rather than independent when computing uncertainty. The release ships room identifiers so this is possible. Compare a score against three reference points, from least to most useful: the random rate, the action-only ceiling for that turn, and the trained-human reference. A score above the ceiling is necessary but not sufficient, because a system can reach it without using the image at all. Do not use the held-out questions for model selection or prompt search. A system tuned against held-out feedback is no longer covered by the exposure accounting in Section 3.4 of the main paper. We follow these rules throughout Section 4.

	Accuracy by rotation					Assumed-ahead rule		Same as no-image	
Model	30°	60°	90°	120°	180°	90°	180°	90°	180°
Gemini 3 Flash	100.0	99.3	40.6	98.9	33.9	92.7	100.0	95.8	100.0
Gemini 3 Pro	95.9	96.7	43.2	95.9	37.5	90.2	96.4	95.8	91.7
Gemini 3 Pro (thinking)	100.0	98.9	41.0	98.9	35.7	91.9	98.2	91.7	95.8
Claude Fable 5	99.3	96.7	34.2	98.5	37.5	98.7	96.4	100.0	100.0
GPT-5.6 Sol (thinking)	97.0	91.9	44.0	94.8	42.9	88.0	91.1	91.7	91.7
Claude Opus 5	95.6	97.0	34.6	97.0	33.9	98.7	100.0	95.8	100.0
Claude Opus 5 (extended)	95.6	97.8	34.2	97.0	33.9	99.1	100.0	100.0	100.0
Claude Sonnet 5	93.3	77.8	49.6	92.9	35.7	81.6	98.2	91.7	100.0
GPT-5.6 Sol	89.6	80.7	44.9	91.4	50.0	88.0	82.1	91.7	79.2
Gemini 3 Flash Lite	77.4	52.6	33.3	50.0	33.9	100.0	100.0	100.0	100.0
Claude Haiku 4.5	49.3	49.6	32.9	50.0	35.7	99.6	30.4	100.0	70.8
GPT-5.1	48.5	46.3	27.4	46.6	30.4	67.5	58.9	83.3	50.0
Mean	**86.8**	**82.1**	**38.3**	**84.3**	**36.8**	**91.3**	**87.6**	**94.8**	**89.9**
Action-only ceiling	50.4	50.4	33.3	50.0	35.7				
Feasible support $\|S(z)\|$	2	2	3	2	3				

Table 6. Task-B accuracy per action cell for all twelve frontier configurations, the full version of the profile in Figure 3 of the main paper. **Assumed-ahead rule** gives the percentage of answers matching $\mathrm{zone}(\delta)$, the prediction from applying the stated turn to a target assumed to lie straight ahead. **Same as no-image** gives the percentage of main-arm answers byte-identical to the same model's no-image answer on the shared 120-item subset, which every configuration now has. The mean falls from 82.1–86.8% at 30°/60°/120° to 38.3% and 36.8% at 90° and 180°, within 5.0 and 1.1 percentage points of the action-only ceiling, the score a system that sees only the stated rotation and turn can already reach.

	Raters	Mean	95% CI	κ
A: identity	9	98.1	[96.2, 99.4]	0.960
B: direction	9	84.6	[77.3, 91.1]	0.681
C: visibility	9	89.6	[83.6, 94.7]	0.820

Table 7. Trained-human performance (%) on 120 items per task. The number we report is mean rater accuracy. Intervals are percentile intervals from a two-way cluster bootstrap over raters and scenes, with 2,000 draws. It resamples both the raters and the scenes, so neither a rater's habits nor a shared room counts as independent evidence. `score_human_responses.py` generates the intervals alongside the accuracies. Fleiss' κ uses semantic answer classes.

Work	Closest overlap with EgoShift	Evaluation emphasis
SCOPE [6]	updating and occlusion, 8 views	consistency; blind and human refs
PCSR [2]	turning in place; limited view	eight perspective tasks
CVT [10]	target view withheld	angle and cycle consistency
MMSI / MindCube [11, 12]	spatial models across views	multi-image; blind and human refs
ReasonMatch / CVS [8, 13]	wide-baseline correspondence	matching, grounding, viewpoint ID
TriView [3]	identity and occlusion, 3 views	complexity scaling; failure modes
Unreal3DSpace [9]	exact synthetic supervision	distribution shift diagnosis
TsT / IBP [1]	shortcuts needing no image	learned attacks; iterative pruning
EgoShift (ours)	**direction after a stated turn**	**a computed, enforced ceiling**

Table 8. The closest benchmarks, and how EgoShift differs. It introduces no new spatial task category, and several of these precede it on viewpoint updating, occlusion, human references and blind controls. The new part is the construction. For each stated turn we compute the highest score a system can reach from that turn alone, and we require the built questions to match it. Shortcuts that use anything else are a separate check, described in Section 5 of the main paper.

7. Benchmark comparison

Tab. 8 tabulates the closest prior and concurrent work along the dimensions on which EgoShift could be confused with it. The table is positioning, not a claim of superiority. EgoShift introduces no new spatial task category, and several of these benchmarks precede it on viewpoint updating, occlusion, human references, blind baselines and embodied correlation.

8. Compute and cost

Evaluation ran in two halves. The twelve paid API configurations consumed 50.7M input and 15.4M output tokens over 47,010 recorded calls, about $363 at the providers' published list prices on the dates of the runs. The no-image control arms account for under $10 of that, because they carry no image tokens. Task-A and Task-C paid runs use frozen 1,192- and 1,200-item subsets rather than the full pools, which is what keeps the paid half at this scale. The open-weight half ran locally on NVIDIA L40S GPUs, across 24 open-weight checkpoints from 2B to 78B parameters. Those runs cover the full Task-A (5,838) and Task-B (1,098) sets, the shared Task-C subset and the control arms. They produced 226,175 of the 273,894 recorded response rows. The full Task-A run is a property of the local runs, not of the reported ranking. We score every Task-A number in this paper on the frozen 1,192-item subset that every model shares, so open-weight and paid systems are compared on identical items. Scene generation, mask replay and the candidate build are separate from evaluation and dominated by AI2-THOR CloudRendering rather than by model

inference. The paid panel is the only part of this work that costs money to reproduce. The audits, the construction and the open-weight panel need time and GPUs but no API budget.

The raised reasoning setting does spend more compute, which is why its flat Task-B result carries weight. GPT-5.6 Sol spends 107 output tokens per question by default and 161 at high effort, for a two-token answer.

9. Datasheet

We follow the structure of Gebru et al. [5]. Answers describe the current candidate, which is not yet a release.

Motivation. We built EgoShift to measure whether a vision–language model updates the direction, identity and visibility of a remembered object after a controlled change in camera pose. We also built it to do so with a construction procedure that bounds a stated image-free shortcut before any model runs. The authors created it as academic research. No external body commissioned it and no funder had any role in its design or in what we report.

Composition. Instances are multiple-choice items built from rendered indoor scenes. Task A holds 13,142 cross-view identity items, Task B 2,130 direction items and Task C 13,275 visibility items, split into held-out, public development and tuning portions. Each item carries one or two 640×480 frames, a question and an answer set. A private record holds the camera poses, object identifiers, instance masks and the geometry that determines the label. Task-B frames share one camera pose per item and differ only in heading. The Task-A and Task-C cameras also translate: they stay between 0.90 and 0.94 metres high, and for Task A within 8° of level. For Task C we record the occluding object by name and depth. Frames are synthetic renderings of furnished interiors and contain no people, no faces and no personal information. Labels come from simulator state, and we check them by replay rather than by annotation, so there is no label noise of the kind that human labelling introduces. The automated checks cannot certify whether an item is clear to a human viewer, which is why the trained panel and the item review exist. Items are not independent, because several can share a physical scene. That is why every interval we report clusters on scenes.

Collection process. Scenes come from ProcTHOR-10k via the `prior` package and are rendered with AI2-THOR CloudRendering [4, 7]. We sample camera poses from the declared range of starting directions, and we choose the target so that exactly one instance of its category is visible above a 30-pixel threshold. The answer follows from the post-action geometry, and we verify it against exact instance masks. We scraped nothing and used no third-party annotation service. Data collection for the model panel ran in August 2026. Human sessions ran between 24 and 27 August 2026, and the panel was frozen at the end of that window.

Preprocessing, cleaning and labelling. We discard items that fail the coverage, visible-pixel, boundary-margin or edge-clipping checks, rather than repairing them. Task-B candidates must meet a 0.80 source silhouette-coverage threshold where defined, hold at least 2,500 visible target pixels, and keep the full target box 15° from a post-turn answer boundary. Task C uses its per-class conditions and a 300-pixel threshold. We fingerprint scene identifiers so that renamed or duplicated scenes collapse to one physical component before splitting, and we assign split roles per component. We keep both the retained items and the audit reports that certify them. We do not keep the raw frames from discarded candidates.

Uses. The intended use is evaluation of spatial updating in vision–language models, including the per-cell reporting this paper argues for. The construction tools are reusable for building other action-conditioned benchmarks with a stated shortcut. Do not use EgoShift as training data for models that will later be evaluated on it. Do not read it as evidence about photorealistic imagery, real sensors or embodied deployment, none of which it contains. Task A is a positive control and carries no difficulty claim, so a score on Task A alone means little.

Distribution. The artifact is public and reachable without contacting the authors. It contains the questions for every split, the development-split answers, a seeded frame sample, the scoring and audit tools, the build reports and the raw per-item model completions. We withhold the held-out *answers*, as is usual for a benchmark with a held-out split, but we distribute the held-out items themselves so any system can be run on them. On acceptance we release the public splits, the construction and audit code, the per-item raw model completions and the exposure ledger. Our annotations and audit outputs go out under CC-BY-4.0, and we redistribute the rendered frames under the upstream Apache-2.0 terms, with attribution and NOTICE preserved. That redistribution rests on an asset audit. We checked all 9,838 staged houses. The 1,465 distinct object assets in them are all AI2-THOR library assets, with no external asset identifiers. The artifact carries Croissant metadata conforming to version 1.0, including the nine Responsible AI fields. Human response files are in no artifact and are not released.

Maintenance. The authors maintain the benchmark. Any change to items, gold labels or prompts produces a new candidate identifier and invalidates results tied to the previous one. Scores are therefore comparable only within a candidate. Errata and superseded candidates are recorded rather than silently replaced. On acceptance the record carries an issue tracker for error reports, and the same channel publishes errata. We cannot name that channel here without

breaking anonymity.

Broader impact. The intended effect is better-calibrated reporting of spatial reasoning. A score on EgoShift does not license a safety claim about a system operating in a real environment. We motivate the work with robots and headsets only to say why the ability matters. The scenes are synthetic interiors from one asset library and do not represent real homes, so a system tuned to score well here may still fail in a real room. The rendered frames contain no people and no personal information.

10. Frozen protocols for checks that have not run

The main paper says the learned attacks are a separate check whose protocol was fixed in advance. That protocol, and the protocols for the other checks we have not yet run, are stated here so that the eventual results cannot be shaped by the outcome.

Residual-shortcut battery. The attack family is fixed before evaluation and runs with scene-grouped cross-validation, so that a classifier cannot succeed by memorising a room appearing on both sides of a fold. It covers global and action-cell majority rules, n-gram and logistic models over the full public text, and a random forest over public metadata fields. It also covers a TsT-style full-text learner [1], option-order and identifier probes, and a zero-shot blind model given every public field but no image. The decision rule is one-sided: the gate fails when the upper 95% scene-cluster confidence bound of the strongest attack exceeds the structural target by more than three absolute points. We fix the margin before evaluation so that nobody can reinterpret the bound once its value is known. Until this battery runs we call no question set clean, and we restrict shortcut claims to the declared action-cell predictor that the structural target does bound.

Presentation controls. Matched conditions compare real-image, no-image, blank, gray, noise, single-view, option-order, free-response, encoding, and motion presentations, with paired effects and cluster-aware intervals reported for every one, including those that go against us. The no-image and single-view conditions have run, and the main paper reports them. The remaining conditions have not run. We hold the reporting language to the statistics available. A non-significant p-value is not evidence of equivalence. Any claim that a manipulation had no meaningful effect therefore requires the paired interval to fall inside the frozen ± 3-point region, with a Holm correction applied within the presentation family.

Action-necessity counterfactuals. Matched items vary only the stated action across correct, absent, sign-flipped, wrong-magnitude, irrelevant-action, and post-action-view conditions, holding the image and target fixed. This is the test that would confirm or refute the assumed-bearing account in the main paper. A model that applies the stated rotation to an assumed straight-ahead bearing should track the sign-flipped and wrong-magnitude conditions exactly, and should stay insensitive to the image. A model that updates the observed bearing should not. Because these have not run, the main paper marks the account as an interpretation of the evidence rather than an established mechanism. These conditions establish whether performance depends on the action as stated. They would not establish that a single latent ability produces every error, and we will not word them that way.

Preregistration and traceability. We content-hash and register the hypotheses, exclusion rules, cluster units, multiplicity families and equivalence margins before we inspect any sealed-split outcome. Every reported row comes from raw predictions tied to the candidate manifest, the prompt and parser hashes, the scorer and the statistical code. A number in either document therefore traces to the bytes that produced it. We exclude results from mixed or superseded builds, which is why the 72.3% leak and the invalidated $494/1{,}140$ seed appear as construction history rather than as measurements.

References

[1] Ellis Brown, Jihan Yang, Shusheng Yang, Rob Fergus, and Saining Xie. Benchmark designers should "train on the test set" to expose exploitable non-visual shortcuts, 2025. 5, 7

[2] Yuangong Chen, Wai Keung Wong, Jiaxing Li, Ioannis Patras, and Xu Zheng. Beyond localization: A comprehensive diagnosis of perspective-conditioned spatial reasoning in MLLMs from omnidirectional images, 2026. 5

[3] Yu-Yang Chen and Lan-Zhe Guo. TriViewBench: Controlled complexity scaling for multi-view structural reasoning in MLLMs, 2026. 5

[4] Matt Deitke, Eli VanderBilt, Alvaro Herrasti, Luca Weihs, Jordi Salvador, Kiana Ehsani, Winson Han, Eric Kolve, Ali Farhadi, Aniruddha Kembhavi, and Roozbeh Mottaghi. ProcTHOR: Large-scale embodied AI using procedural generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 6

[5] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. 6

[6] Yoonji Kim, Jieun Kim, Yujin Jeong, and Sung-Bae Cho. Diagnosing spatial consistency across perspectives and viewpoints in large vision-language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational*

Linguistics (Volume 1: Long Papers), pages 32803–32827, 2026. 5

[7] Eric Kolve, Roozbeh Mottaghi, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Matt Deitke, Kiana Ehsani, Daniel Gordon, Yuke Zhu, Aniruddha Kembhavi, Abhinav Gupta, and Ali Farhadi. AI2-THOR: An interactive 3D environment for visual AI. *arXiv preprint arXiv:1712.05474*, 2017. 6

[8] Ruixun Liu, Lingyu Zhang, Lanxuan Xue, Kaiyu Li, Bowen Fu, and Xiangyong Cao. CVSBench: A comprehensive benchmark for cross-view spatial reasoning and dreaming, 2026. 5

[9] Wufei Ma, Sky Cen, Jianzhi Shen, Rex Lee, León Begiristain, Yan Zhuang, Jiawei Peng, Zhifei Yu, Tianao Song, Xinyuan Qi, Tianmin Shu, Adam Kortylewski, and Alan Yuille. UnrealSpace: Analyzing spatial understanding and reasoning in controllable simulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, pages 9716–9725, 2026. 5

[10] Shanmukha Vellamcheti, Uday Kiran Kothapalli, Disharee Bhowmick, and Sathyanarayanan N. Aakur. CVT-Bench: Probing spatial-state integrity through counterfactual viewpoint transformations. *arXiv preprint arXiv:2603.21114*, 2026. Version 2, 14 August 2026. 5

[11] Sihan Yang, Runsen Xu, Yiman Xie, Sizhe Yang, Mo Li, Jingli Lin, Chenming Zhu, Xiaochen Chen, Haodong Duan, Xiangyu Yue, Dahua Lin, Tai Wang, and Jiangmiao Pang. MMSI-Bench: A benchmark for multi-image spatial intelligence. In *ICLR*, 2026. arXiv:2505.23764. 5

[12] Baiqiao Yin, Qineng Wang, Pingyue Zhang, Jianshu Zhang, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, Han Liu, Ranjay Krishna, Saining Xie, Manling Li, Jiajun Wu, and Li Fei-Fei. MindCube: Spatial mental modeling from limited views. In *ICLR*, 2026. arXiv:2506.21458. 5

[13] Hao Zhong, Muzhi Zhu, Shenyan Zeng, Anzhou Li, Cong Chen, Hua Geng, Duochao Shi, Wentao Ye, Tao Lin, Hao Chen, and Chunhua Shen. Eliciting complex spatial reasoning in MLLMs through wide-baseline matching, 2026. 5